Semantic Analysis of Domain Data for Proactive Identification of Cybersecurity Indicators

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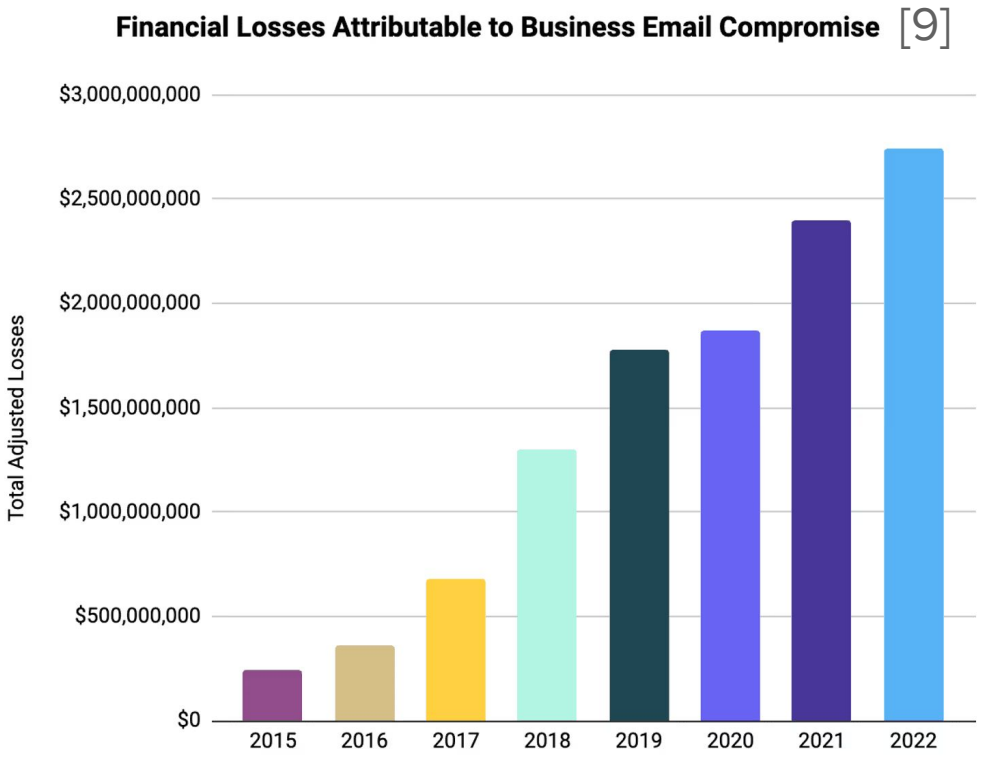
Introduction to Our Topic

- Phishing and Spam **wastes resources** and **harms end users**
- Threats rise while **detection stagnates**
- AI and **Data Science abilities have evolved** since early Spam filters were made
- Many avenues exist to **expand the old guard** of solutions



Overall Literature Overview

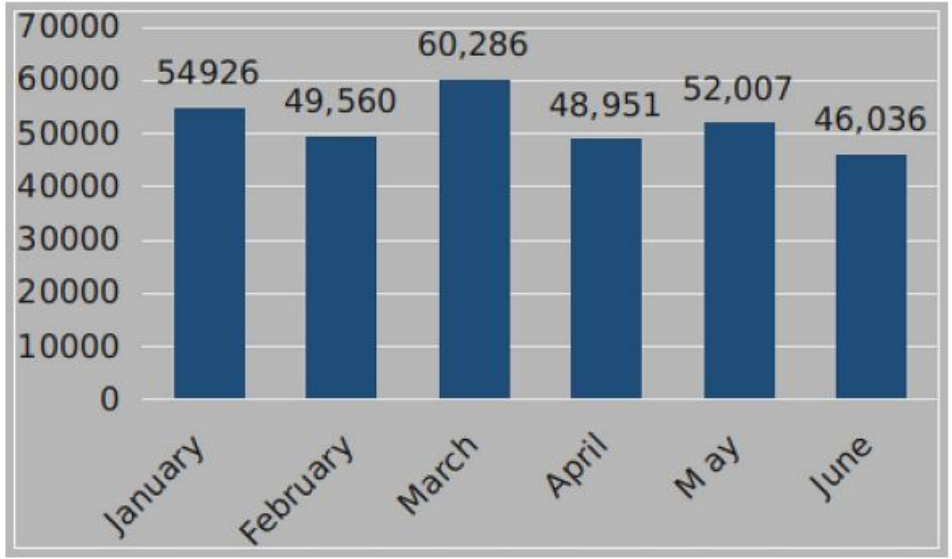
The primary categories of interest for our topic were:

- **Reactive Detection** of Attacks
- **Proactive Data** Gathering
- **Proactive Prevention** of Attacks
- **Psychological Studies**

Review of Key Papers - Reactive Detection of Attacks

- Major contributors include 2020's *A* ***semantic-based classification*** *approach for an enhanced spam detection* [5] and *Detecting Phishing SMS Based on* ***Multiple Correlation Algorithms*** [6]

Fig. 1 Number of unique phishing Web sites detected [6]



Review of Key Papers - Proactive Data Gathering

- 2019's *iMCircle: **Automatic Mining of Indicators** of Compromise...* [7] and 2023's *Phishing or Not Phishing? A **Survey**...* [8] both push beyond reactionary remediation

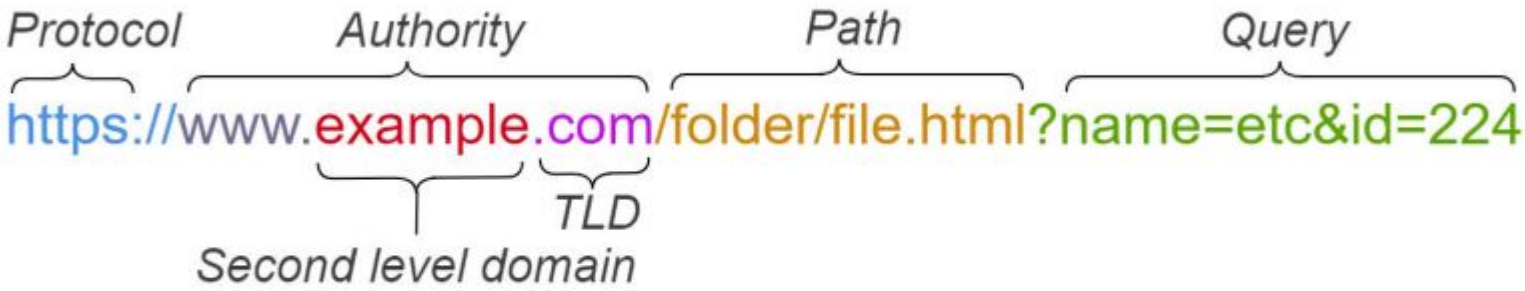


FIGURE 4. Example of a URL with its components. [8]

Review of Key Papers - Proactive Prevention of Attacks

- 2025 brought us *Phish Fighter:* ***Self Updating Machine Learning***... [1] and *APOLLO: A* ***GPT-based tool***... [4] as preventative and semantic tools following the boom of LLMs
- Proactive remediation costs more, but the cost of a single attack can be **many orders of magnitude more** [9]
- Despite this, the evolution of proactive capabilities have **lowered cost while also increasing effectiveness**

Review of Key Papers - Psychological Reviews

[3] **TABLE 5** Correlations with actions selected for phishing emails.

Measure	Follow request	Investigate	Flag	Ignore/delete
d'	−0.45**	−0.16	0.17*	0.33**
c	−0.03	−0.10	0.00	0.09
Hits	−0.42**	−0.12	0.17*	0.29**
FAs	0.45**	0.18*	−0.18	−0.34**
RT	−0.32**	−0.01	0.17*	0.13
Confidence	−0.20*	−0.39**	0.18*	0.32**
PAQ	0.10	−0.18*	0.10	−0.01
Age	−0.33**	−0.14	0.11	0.27**
Big 5-E	0.08	−0.07	−0.08	0.04
Big 5-A	−0.43**	−0.30**	0.18*	0.42**
Big 5-C	−0.37**	−0.30**	0.19*	0.37**
Big 5-N	0.29**	0.21*	−0.13	−0.28**
Big 5-O	−0.05	−0.19*	0.03	0.16*
Impulsivity	0.47**	0.28**	−0.22**	−0.41**
Curiosity-SC	0.27**	0.06	−0.10	−0.18
Curiosity-DS	0.22**	0.03	−0.12	−0.10
Curiosity-ST	0.36**	0.21**	−0.10	−0.35**

*$p < .05$; **$p < .01$.

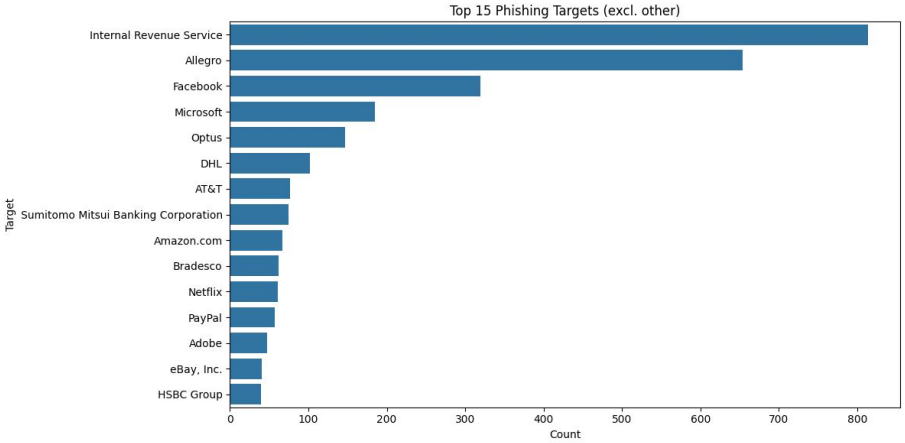
- And in a more foundational sense, the 2022-2023 papers of *An **Effective Cybersecurity Awareness Training** Model...* [2] and *Which phish is captured in the net?...* [3] show **surveys** on how the previously mentioned processes **actually affect people**

Primary Research Gaps Identified

- The Zero-Day Challenge
  - Current systems **struggle with novel threats**.
  - Detection accuracy degrades rapidly without **prior threat data**.
- Evolution of ML Capabilities
  - Shift from manual feature engineering to **semantic LLM reasoning**.
- Empowering the End User
  - Focus on easily interpretable outputs to **assist human decision-making**

How Our Project Will Build Upon Existing Work

- **Domain Focused Semantic Analysis**
- Tranco Top 50K Domains & Phishtank Verified Phishing URLs [10] [11]
- Semantic Understanding via **Gemini Text Embeddings** & Comparisons via **Cosine Similarity**
- Neither dataset is infallible
- Focus on **Actionable Intel**




```
                DATA SHAPE
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Avg PhishTank URL length      : 71.2
Avg PhishTank host length     : 20.2
Avg Tranco domain length      : 12.3
Median PhishTank URL length   : 38
Median Tranco domain length   : 12
Mean hyphens — PhishTank      : 0.93
Mean hyphens — Tranco         : 0.09
Mean digit ratio — PhishTank  : 0.1023
Mean digit ratio — Tranco     : 0.0124
PhishTank unique hosters      : 8,271
Tranco unique domains         : 50,000
```
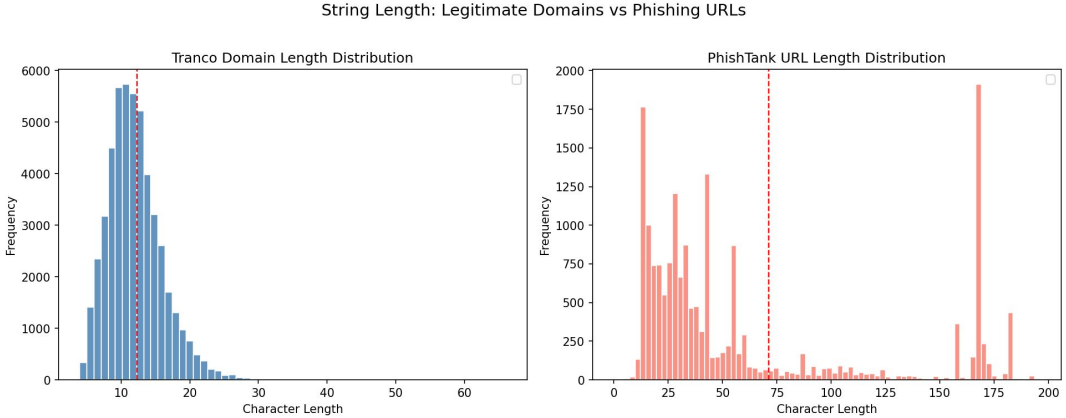
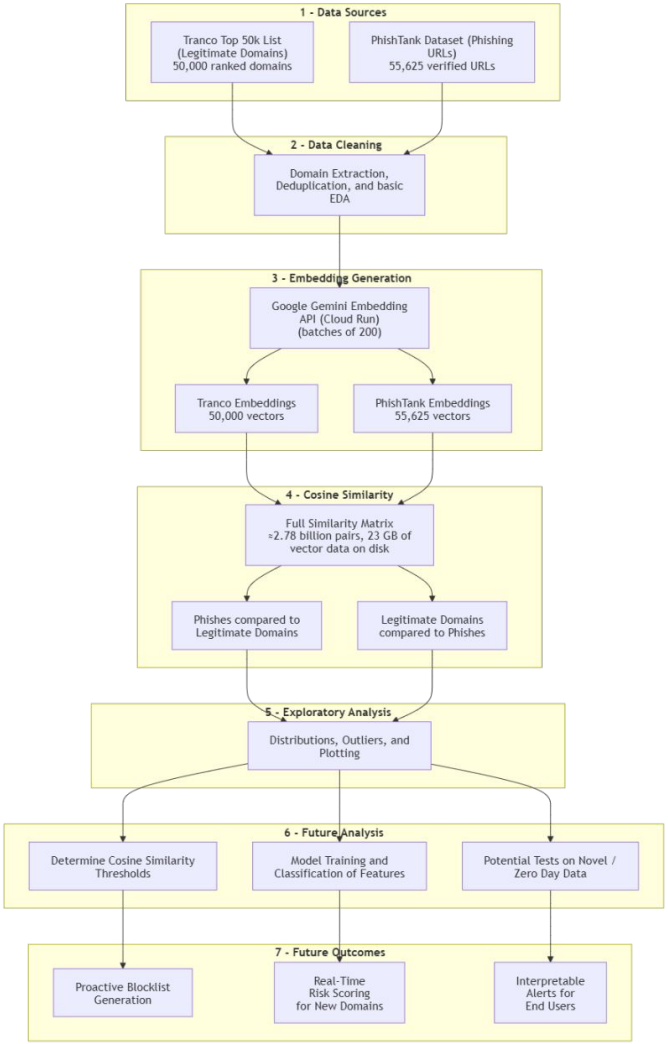



Summary & Conclusion

- Semantic Analysis is likely to become **instrumental in fighting** future attacks
- Embeddings are becoming more and **more reasonable** to be done locally
- Plenty of unforeseen data features and insights lie in the **semantic metadata** of the average domain



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